AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION - REVISION 5.0
Document ID: HLD-POWERTRAIN-2026-V5.0
AUTOSAR Release: Adaptive Platform 20-11 & Classic 4.4.0 (Hybrid Gateway)
Target ECU: CentralPowertrainController (HPC_ECU), BatteryManagementSystem (BMS_ECU)

1. Executive Summary & System Overview
This revision 5.0 updates the EV Powertrain architecture to support a Centralized High-Performance Computing (HPC) controller running Adaptive AUTOSAR with SOME/IP over Automotive Ethernet, alongside Classic AUTOSAR ECUs.

2. Software Architecture Scope
Replaces discrete CAN-FD signals with service-oriented communication (SOME/IP) for powertrain telemetry and incorporates predictive energy management.

3. Software Components (SWC) Inventory

3.1 TorqueArbitration_SWC
- Type: Adaptive Application (ara::com Service)
- Purpose: Service-oriented torque arbitration supporting autonomous driving ADAS inputs and driver pedals.
- Interface: TorqueService (SOME/IP skeleton/proxy)

3.2 PredictiveEnergyManager_SWC
- Type: Adaptive Application
- Purpose: Uses GPS terrain and route look-ahead to optimize battery state-of-charge.

3.3 InverterMotorControl_SWC
- Type: Classic AUTOSAR SWC
- Purpose: Direct actuator control over CAN-FD gateway.